Akash Dubey

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Education

Rutgers University Honors College, School of Arts and Sciences New Brunswick, NJ
Bachelor of Science in Computer Science and Mathematics *Anticipated May 2027*

- **GPA**: 3.95/4.0 | SAT: 1580
- **Relevant Coursework**: Stochastic Processes, Math Theory of Probability, Honors Calculus III & IV, Algorithms, Data Structures, Systems Programming, Computer Architecture, Tensor Networks.
- **Academic Programs & Research**: Directed Research Reading Program on Symmetric Functions (representation theory for combinatorial optimization); Physics Learning Theory Seminar (statistical mechanics in high-dimensional learning), Dean's List.
- **Activities**: Quantitative Finance Club, Road to Silicon Valley Program (Cohort 6), ESL Instructor, Pickleball.

Engineering Experience

New York Life Insurance New York, NY
Software Engineer Intern, TDAV Team *May – Aug 2026*

- Built an AI text and voice agent that automates insurance claims processing for clients and augments the workflows of 12,000 Customer Service Representatives, deployed on AWS infrastructure with secure access to internal services, knowledge, and forms.
- Constructed an enterprise knowledge base from 13,000 documents compiled across SharePoint, Public Site, Confluence, Jira, and the Intranet, consolidating fragmented business logic into one retrieval layer for AI and human users.
- Engineered multi-agent systems on NYL's cloud to automate analyst-level work across AEM web development, Pega code, Tungsten document processing, and mainframe modernization.

Samaritan Scout Cranford, NJ
Full Stack Engineer Intern *May 2023 – Aug 2025*

- Developed an agentic scraping system leveraging OpenAI and Gemini structured outputs to autonomously extract structured data from over 100,000 websites, reducing manual data-entry work by over 90%.
- Built a full-stack search engine using React, TypeScript, and PostgreSQL to index 35,000+ records, implementing semantic search with hallucination filters to ensure accurate, auditable matching.

Lykke San Francisco, CA
Founding Software Engineer *Sep 2025 – Present*

- Built core AI workflows that ingest documents, notes, and source materials into scoped knowledge bases for contextual chat and automated content generation.
- Engineered adaptive RAG pipelines across MongoDB, Zilliz/Milvus, S3, and SSE streaming, including source inference, citation events, token-budgeting for large document sets, and fallback paths for extracted text vs. full-file reasoning.
- Created a distributed wiki-generation engine that crawls sources, synthesizes structured sections with citations, embeds content for retrieval, and runs tenant-isolated background builds via SQS, MongoDB leases, Redis rate limiting, and AWS ECS.

Scarlet Sync New Brunswick, NJ
Founder *Jan 2025 – Present*

- Built a planning platform with thousands of users, using AI-assisted parsing of unstructured data from legacy services to power degree, schedule, and discovery workflows.
- Architected a robust data ingestion pipeline in Python with LLMs to reverse-engineer legacy data, indexing over 2,500 classes and 500+ programs for real-time querying.
- Developed a production AI assistant with streaming tool-calling and grounded retrieval, enabling automated planning across Web, ChatGPT, Claude, and iOS.

Research

Rutgers Economics Labs New Brunswick, NJ
President (formerly Research Director, Team Lead) *Oct 2024 – Present*

- Lead quantitative research projects for partners including NJDOL, NJDEP, NJBPU, Federal HUD, and the U.S. Census Bureau; apply econometric models in Python/R to analyze large datasets and project economic trends.
- Architected RAIL (Rutgers Agentic Intelligence Labs), an AI-automated economic research pipeline that collects, organizes, analyzes, and synthesizes economic data to derive insights using data ontologies and coding-agent workflows.
- Previously led a team of five researchers evaluating the efficacy of the Urban Enterprise Zone program for the NJDCA, applying econometric methods in R and Python to complex census datasets.

Algorithmic Robotics and Control Lab in Rutgers's Dept. of CS New Brunswick, NJ
Research Assistant (Advisor: Prof. Jingjin Yu) *Jan 2026 – Present*

- Implemented and validated CUDA versions of pRRTC and FCIT*, parallelizing tree/graph search and collision checking for high-dimensional ASAO robot motion planning across 500 benchmark problems.
- Developing a graph-based search policy (GaP) for dual-arm task-and-motion planning (TAMP), learning from successful and failed planner traces to rank symbolic action skeletons and predict geometric feasibility.

Selected Technical Projects

Knowledge Base and AI Agent Workflow Manager | *Data Ontology, Workflows, Coding Agents*

- Built KRAIL (pypi), a local-first, repo-backed knowledge runtime that structures raw notes into ontology-aware topic pages, source ledgers, claim records, verification runs, and artifact lineage for evidence-grounded agent memory.
- Implemented MCP/CLI and agent infrastructure for auditable coding-agent workflows (software factories, auto-research, and company-brain use cases), including runner dispatch, dry-run work orders, session logs, role prompts/checklists, workflow validation, and hybrid keyword-graph-vector retrieval.

Local Coworker: Native macOS Computer-Use Agent | *Swift, MCP, Gemini Vision* (private code)

- Built a native desktop agent capable of autonomous computer use by integrating Gemini Vision with macOS Accessibility APIs, AppleScript, and JXA.
- Implemented the Model Context Protocol (MCP) to orchestrate specialized sub-agents (UI Interaction, Shell, App Research) for planning and executing complex multi-app workflows.

SLM Optimization & Post-Training (GSM8k) | *PyTorch, HPC*

- Fine-tuned small language models (SLMs) on the GSM8k math reasoning dataset to study how model size and architecture affect quantitative reasoning capabilities.
- Deployed models on the Rutgers High Performance Computing (Amarel) cluster, utilizing quantization and structural pruning to maximize inference throughput on constrained compute.

Awards & Skills

Awards: 8090 AI Top Coder (5th), Rutgers Shark Tank (2nd and 3rd), ASA Fall Data Challenge (Top 3), NSF I-Corps.
Languages: Python, SQL, TypeScript, C++, Java, R, Bash, Git
Technologies: Excel, PowerPoint, Python/R, Forecasting, Quantitative Analysis, LLMs, Data Pipelines, Document Extraction, AWS, PostgreSQL, Jira, Pandas, PyTorch